Arka Mukherjee

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- **Research Focus**: LLM reasoning, AI agents, and evaluation science

Education

- **Kalinga Institute of Industrial Technology (KIIT)** — Bhubaneswar, India
  Bachelor of Technology in Computer Science; Cumulative GPA: 9.74/10.0 — *July 2023 - Apr 2027 (Expected)*
  - **Relevant Courses**: Agentic AI, Probability & Statistics, Data Analytics, Machine Learning, AI, Algorithms, Data Mining, Human-Computer Interaction (Coursera)

Experience

- **Advanced Micro Devices (AMD) - Dr. S Pavankumar Dubagunta** — Bengaluru, India
  Research Intern — *September 2026 - Present*
  - **LLM Inference**: Contributing to lossless LLM decoding techniques and agentic model optimization frameworks to speed up local AI on AMD GPUs.
- **University of Southern California - Dr. Maja Matarić** — Los Angeles, CA
  IUSSTF-Viterbi Summer Research Intern — *May - August 2026*
  - **Long Video Comprehension**: Built LazySloth, a fast video understanding method that improved retrieval efficiency by up to 8.3× over existing pipelines such as VideoLucy and WorldMM, while retaining task accuracy. Contributed to lab infrastructure on LLM inference. *(Submitted to AAAI 2027)*
- **IIT Bhubaneswar - Dr. Shreya Ghosh** — Bhubaneswar, India
  Undergraduate Research Fellow (Funded) — *December 2024 - Present*
  - **Multimodal STEM reasoning**: Developed mmJEE-Eval, a multimodal reasoning benchmark of 1,460 STEM problems to combat saturation (Qwen 3 VL 235B scored 45% vs 85% on Math Vista). Evaluated 17 VLMs and discovered metacognitive barriers: VLMs detect 53% of errors but correct only 3.5%. *(IJCNLP-AACL 2025)*
  - **Multimodal Cultural Competence**: Created the first systematic **evaluation framework for VLM cultural competence** via story generation on 5 VLMs with a novel evaluation metric. *(Oral @ ICCV 2025 ASI workshop)*
  - **Integrated Capability Evaluation (ICE)**: Proposed **ICE**, a new evaluation paradigm unlike benchmarks and leaderboards that models real-world tasks. Engineered CoTDL-mini-SWE-agent, a novel coding harness to improve frontier agentic terminal coding performance.
  - **Verifiable Mathematical Reasoning**: Built **MathTypes**, a typed language that provides deterministic step-level feedback for natural language math reasoning. Paired with a compile-verify-correct loop, it increases frontier LLM capabilities in difficult math exams by 9-21%. *Submitted to ICLR 2027*
- **Carnegie Mellon University - Dr. Min Xu** — Remote
  Research Intern — *January - May 2026*
  - **Reinforcement Learning**: Applied GRPO, DPO, and PPO to study RL generalization in a low-sampling decoding task (64-token action space). Improved VRAM efficiency of the code to iterate GRPO on 800k+ datapoints.
- **IIT Ropar - Dr. Sudarshan Iyengar** — Ropar, India
  Summer Research Fellow (IASc-INSA-NASI) — *May - July 2025*
  - **VLM Benchmarks**: Developed **EduVLM-Bench** for STEM prerequisite detection and evaluated OSS LLMs. The top model, Gemma3 27B, achieved 38.5% accuracy.

Publications

1. Mukherjee, A., Roy, S., Trivedi, K., & Ghosh, S. (2026). GeoAgent: Evaluating VLM Geolocalization Through Embodied Navigation (EMNLP 2026 Findings, 14.3% acceptance rate)
2. Mukherjee, A., & Ghosh, S. (2025). mmJEE-Eval: A Bilingual Multimodal Benchmark for Evaluating Scientific Reasoning in Vision-Language Models (IJCNLP-AACL 2025 Findings, 12% acceptance rate)
3. Mukherjee, A. & Ghosh, S. (2025). Toward Socially Aware Vision-Language Models: Evaluating Cultural Competence Through Multimodal Story Generation (ICCV 2025 ASI workshop - Oral)

Projects

- **Automated VLM reasoning dataset generator**: Curated 10k samples for SFTing VLMs. Our method achieved the best VMCBench results, beating Walton Cold Start and MM MathInstruct. *(Ranked #6 in NeurIPS DCVLR challenge)*

Achievements/Extracurriculars

- **Awards**: **Singapore AI Safety Hub (SASH) Frontier AI Security Training (FAST) fellow** (fully-funded, 2.8% acceptance rate), **Google DeepMind APAC Research Symposium** student presenter, **KIIT Merit Scholar** (Dean's list) for 6 semesters, **Amgen Scholars** Summer Research Program (declined), **Aalto Science Institute (AScI)** Summer Research Program in Finland (declined)
- **Outreach**: Published articles (10M+ reads) and hardware reviews in collaboration with Nvidia, AMD, and ASUS.
- **Nvidia GPU Sponsorship**: Received RTX 5090, 5080, and 5070 GPUs from Nvidia India for AI research.
- **Travel Awards & Grants**: D&I Subsidy award at IJCNLP-AACL'25 ($750), AI Grants India (API keys and credits)

Technologies and Skills

- **Languages**: Python (advanced), Java, C, SQL
- **ML/NLP Frameworks**: PyTorch, HF Transformers, Datasets & TRL, Unsloth, LM Studio, Docker, Agentic AI (LangChain, Playwright, mini-SWE-agent)